

Direction 1: We shall prove $p \rightarrow (q \vee r)$ is True by a direct proof

Let p be True. $\therefore m^2 = n^2 \Rightarrow m^2 - n^2 = 0 \Rightarrow (m-n)(m+n) = 0$

Atleast one of $(m-n)$ or $(m+n)$ is 0.

Let, $m-n=0 \therefore m=n$. So q is True. By defn. of OR, $(q \vee r)$ is True.

Let $m+n=0 \therefore m=-n$. So r is True. By defn. of OR, $(q \vee r)$ is True.

By defn. of conditional, $p \rightarrow (q \vee r)$ is True.

Direction 2: We shall prove $(q \vee r) \rightarrow p$ is True by a direct proof

Let $(q \vee r)$ be True. By defn. of OR, the following 3 cases arise:

1. q is True. r is False. $\therefore m=n$ and $m \neq -n$
 $\therefore m^2 = m \cdot m = n \cdot n = n^2$. So p is True

2. q is False. r is True. $\therefore m \neq n$ and $m = -n$
 $\therefore m^2 = m \cdot m = (-n) \cdot (-n) = n^2$. So p is True

3. q is True. r is True. $\therefore m=n$ and $m=-n$
 $\therefore n = -n \Rightarrow n+n=0 \Rightarrow 2n=0 \Rightarrow n=0$. $\therefore m=0$

So, $m^2 = m \cdot m = 0 \cdot 0 = 0 = 0 \cdot 0 = n \cdot n = n^2$. So p is True

By Defn. of conditional, $(q \vee r) \rightarrow p$ is True.

$\therefore p \leftrightarrow (q \vee r)$ is True. So $m^2 = n^2$ if and only if $m=n$ or $m=-n$. $\square$

31) Prove or disprove that if m and n are integers such that $mn=1$, then either $m=1$ and $n=1$, or else $m=-1$ and $n=-1$.

Solution: $p: mn=1$ $q: m$ is an integer $r: n$ is an integer
 $s: m=1$ $t: n=1$ $u: m=-1$ $v: n=-1$

We have to prove that $(p \wedge q \wedge r) \rightarrow ((s \wedge t) \vee (u \wedge v))$ is True.

We shall give a proof by contraposition.